

Democratizing Data for Student Success

A Strategic Proposal for Open Education Analytics (OEA) Implementation

Turning 10 years of education data into
immediate student interventions.



PROPOSAL FOR DISTRICT LEADERSHIP & CABINET

Executive Summary: The Smart Pilot



THE PROBLEM

Education data is siloed and prohibitively expensive to analyze. Traditional enterprise tools cost ~\$188k/year and rely on lagging indicators that arrive too late to help students.



THE SOLUTION

Open Education Analytics (OEA)—an open-source framework utilizing the modern OSS stack (DuckDB, dbt, Metabase). This reduces costs by ~90% while increasing processing speed.



THE GOAL

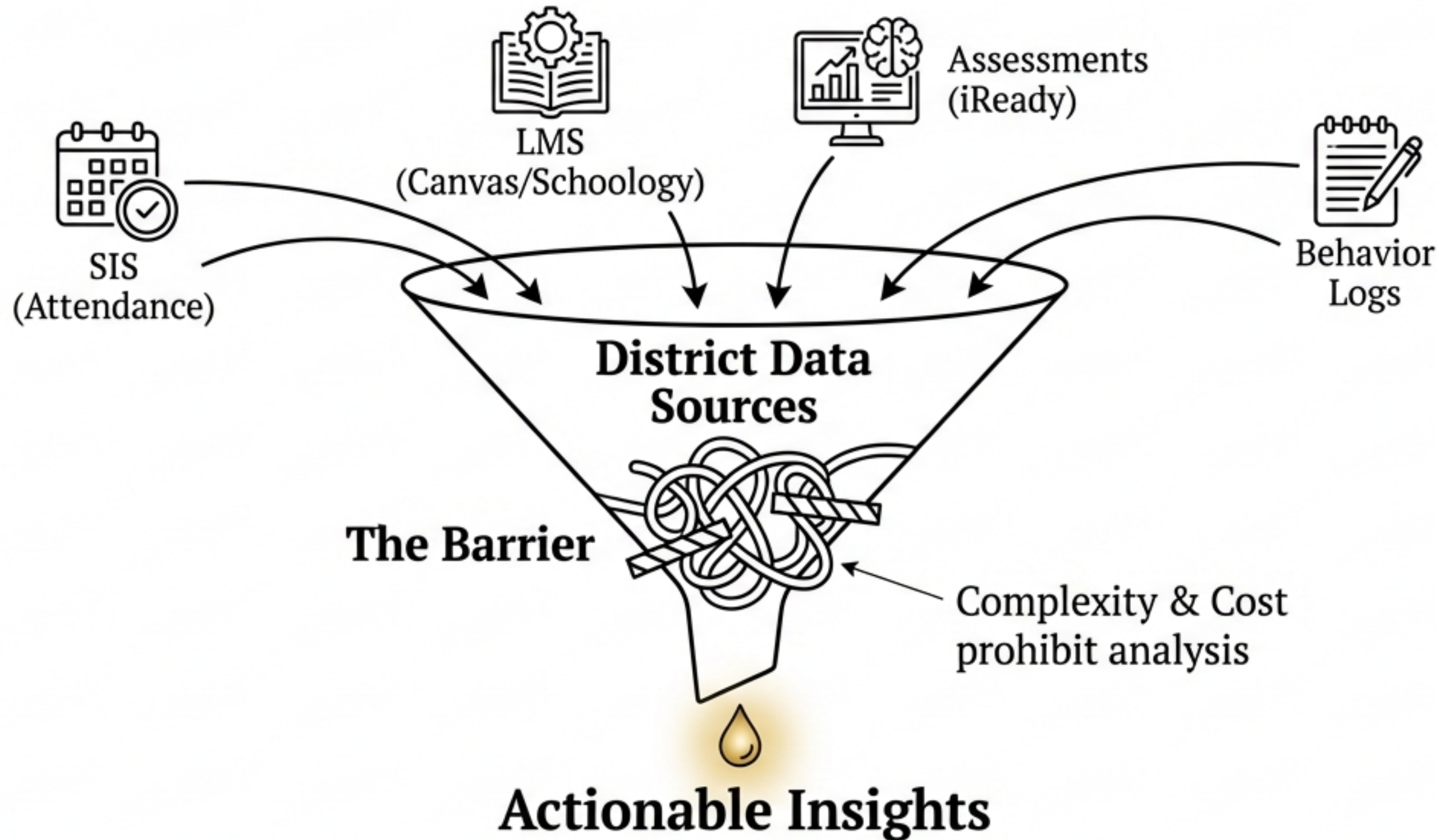
A focused 3-month pilot to predict and prevent Chronic Absenteeism. Moving from reporting on who was absent to predicting who *will be* absent.



THE ASK

Approval for a \$12k-\$15k infrastructure setup budget to prove value before scaling. A low-risk learning opportunity with high potential ROI.

We Have More Data Than Ever, But Use Less Than We Could



35%

Projected job growth for data scientists, creating a talent gap education cannot fill.

Current State: Relying on lagging indicators (quarterly reports) instead of leading indicators (daily patterns)

Why Haven't We Solved This Yet?

Three structural barriers have historically prevented districts from adopting advanced analytics.



Cost

Enterprise analytics tools (Azure/Synapse) typically cost ~\$188k/year in licensing and heavy cloud infrastructure.



Complexity

Traditional systems require teams of specialized data engineers and data scientists that districts cannot hire or retain.



Privacy

Legitimate fears over data vendor lock-in, PII exposure, and sending student data to third-party “black boxes”.

Until now, advanced analytics was a luxury reserved for the largest, wealthiest districts.

The Breakthrough: Open Education Analytics (OEA)

Enterprise-Grade Analytics. Zero Licensing Costs.

Traditional Model

- Proprietary Software
- High Annual Licensing Fees
- Vendor Lock-in
- Generic Business Tools
- Data Sent to 3rd Party

The OEA Model

- Open Source Framework
- Zero Licensing Costs
- Community Driven Modules
- Education-Specific Packages
- Data Stays in District

Key Concept: We don't buy the capability; we adopt the framework.
OEA uses pre-built 'Modules' and 'Packages' specifically designed for schools.

The Architecture: A Three-Layer Engine

Layer 3: Packages
Solution Kits

Layer 2: Modules
Reusable Pipelines

Layer 1: Schemas
Standardizing Data



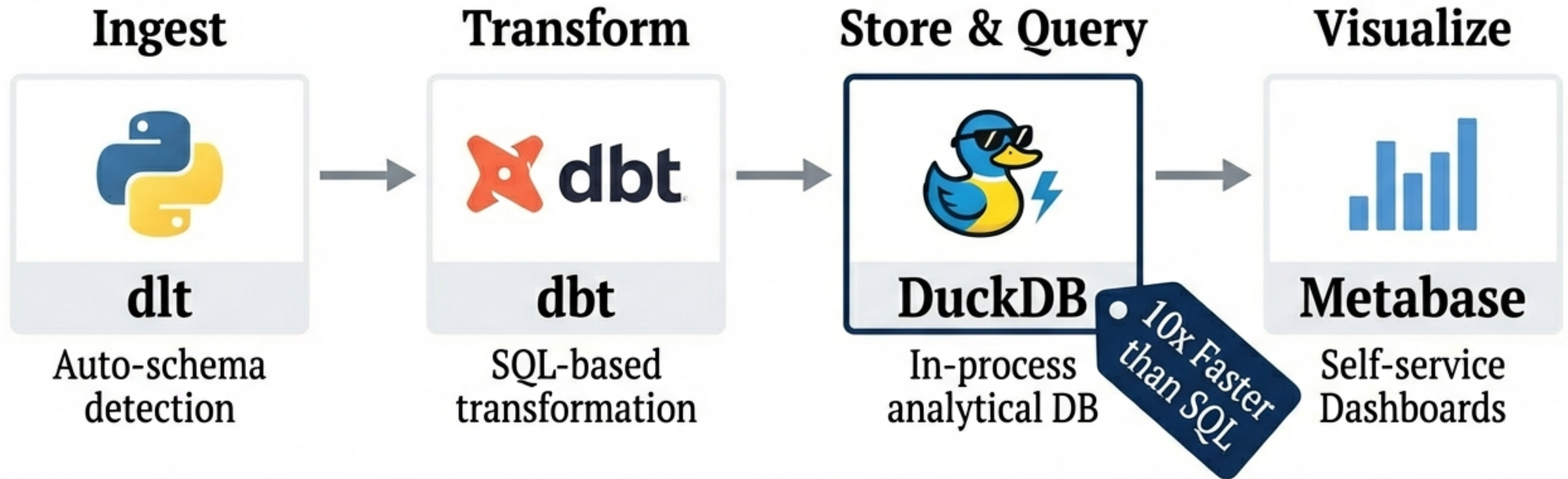
The end-products:
“Chronic Absenteeism”,
“Student Well-Being”,
“College Readiness”.

Pre-built connectors that
ingest and refine data
automatically.

Ensures Canvas, Skyward,
and iReady speak the
same language.

We don't build from scratch; we assemble proven blocks shared by the education community.

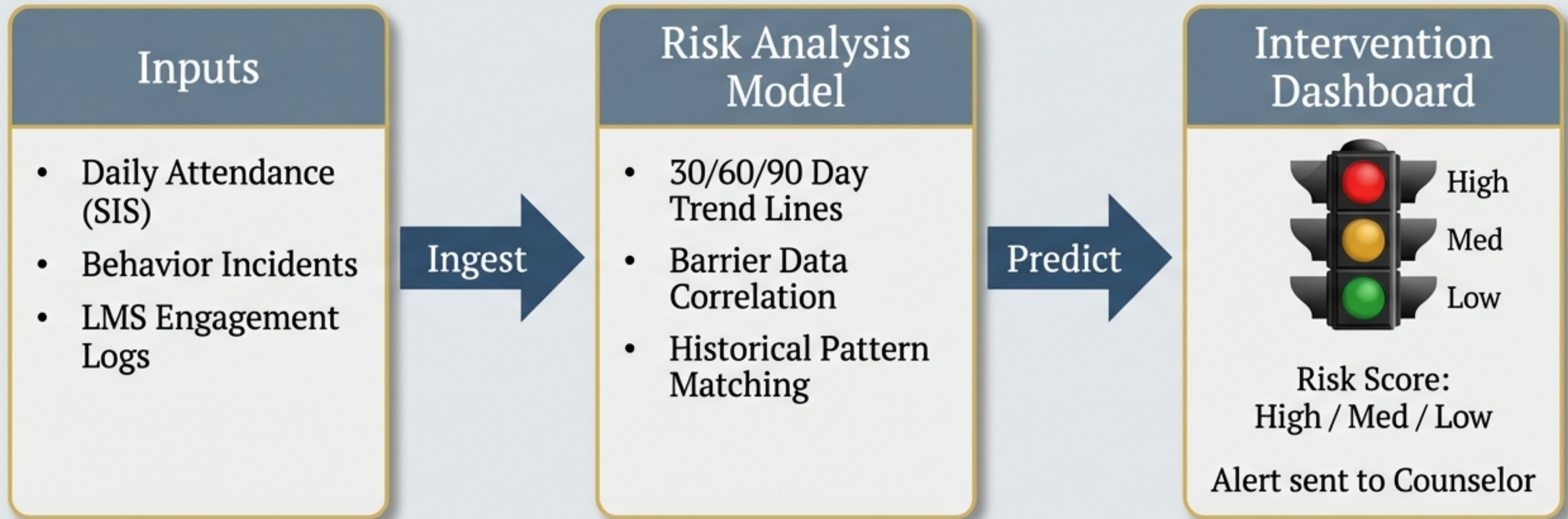
The Technology Stack: Modern, Fast, Secure



Analysis that took minutes on old servers takes **seconds** on **DuckDB**.

Priority Use Case: Chronic Absenteeism

Moving from Reporting to Predicting



“Reporting tells us who was absent. OEA tells us who is about to drop off.”

Impact: Proven Results in Districts Like Ours

“The question isn’t whether OEA works—other districts prove it does. The question is: Are we ready to put our data to work?”

Rural District Scenario (1,200 Students)

Limited IT staff, tight budget, spreadsheet-based tracking.

Graduation Rate



Chronic Absenteeism



ROI

Financial Impact: System cost \$12k.
Avoided hiring \$60k analyst.

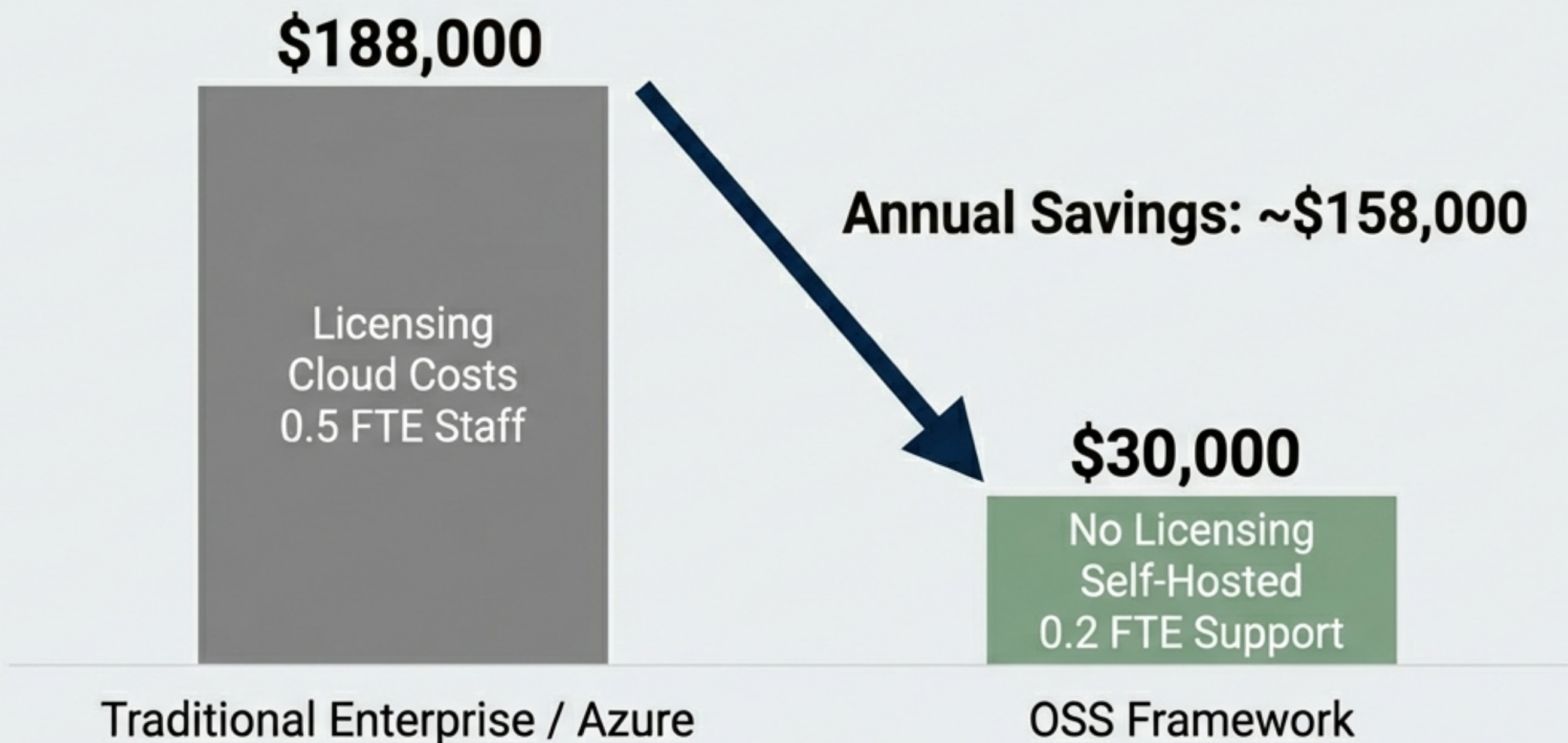
Safety By Design: Privacy & FERPA Compliance



Full compliance with FERPA data retention and privacy standards.

The Financial Case is Undeniable

Total Cost of Ownership (Annual)



Pilot infrastructure cost is only \$12k-\$15k.

A 90-Day Path to Insight



**We are not asking for a multi-year contract.
We are asking for 12 weeks.**

How We Will Measure Success

Technical Success

Live dashboard with automated daily refresh (2 AM UTC).

Adoption Success

20%+ of target educators and counselors using the system weekly.

Outcome Success

At least 5 data-driven interventions deployed during the pilot phase.

Financial Success

Implementation stays strictly within the \$15k setup budget.

Anticipating Your Questions

Is this too good to be true?

It is well-engineered open source. Like Linux or WordPress, you pay for implementation, not intellectual property.

What if we lack technical staff?

The pilot covers initial engineering. Ongoing operations are low-maintenance (0.2 FTE) thanks to automation.

What if the pilot fails?

We lose 3 months and \$12k, not 3 years and \$500k. This is a low-risk learning opportunity.

Our Recommendation

Approve a **3-Month Pilot** focused on **Chronic Absenteeism**.

Resource Request: **\$12k-\$15k** for infrastructure setup & initial engineering support.

Let's turn our data into the safety net our students deserve.

1. Approve Budget. 2. Assign Project Lead. 3. Kickoff Meeting: [Date Placeholder].